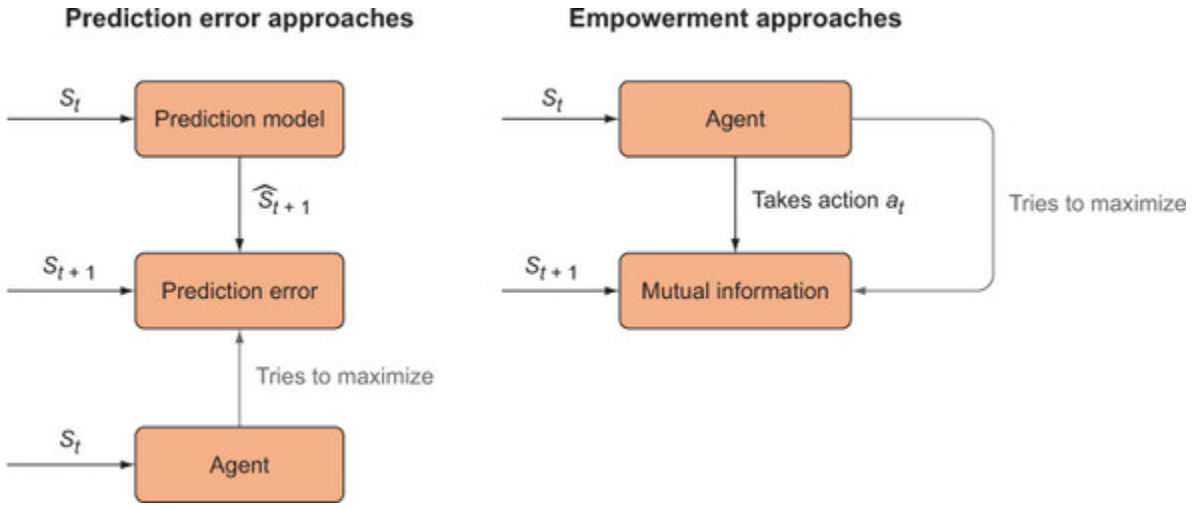
Learning" by Shakir Mohamed and Danilo Jimenez Rezende (2015). We can make the informal statement about maximizing control over the environment into a precise mathematical statement (which we will only approximate here).

Figure 8.21. The two main approaches for solving the sparse reward problem with curiosity-like methods are prediction error methods, like the one we used in this chapter, and empowerment methods. Rather than trying to maximize the prediction error between a given state and the next predicted state, empowerment methods aim to maximize the mutual information (MI) between the agent's actions and the next states. If the MI between the agent's action and the next state is high, that means the agent has a high level of control (or power) over the resulting next states (i.e., if you know which action the agent took, you can predict the next state well). This incentivizes the agent to learn how to maximally control the environment.



The premise relies on the quantity called *mutual information* (MI). We will not define it mathematically here, but informally, MI measures how much information is shared between two sources of data called *random variables* (because usually we deal with data that has some amount of randomness or uncertainty). Another less tautological definition is that MI measures how much your uncertainty about one quantity, *x*, is reduced given another quantity, *y*.

Information theory was first developed with real-world communication problems in mind, where one problem is how to best encode messages across a possibly noisy communication channel so that the received message is the least corrupted (figure 8.22). Suppose we have an original message *x* that we want to send across a noisy communication line (e.g.,